SMART INDIA HACKATHON 2026

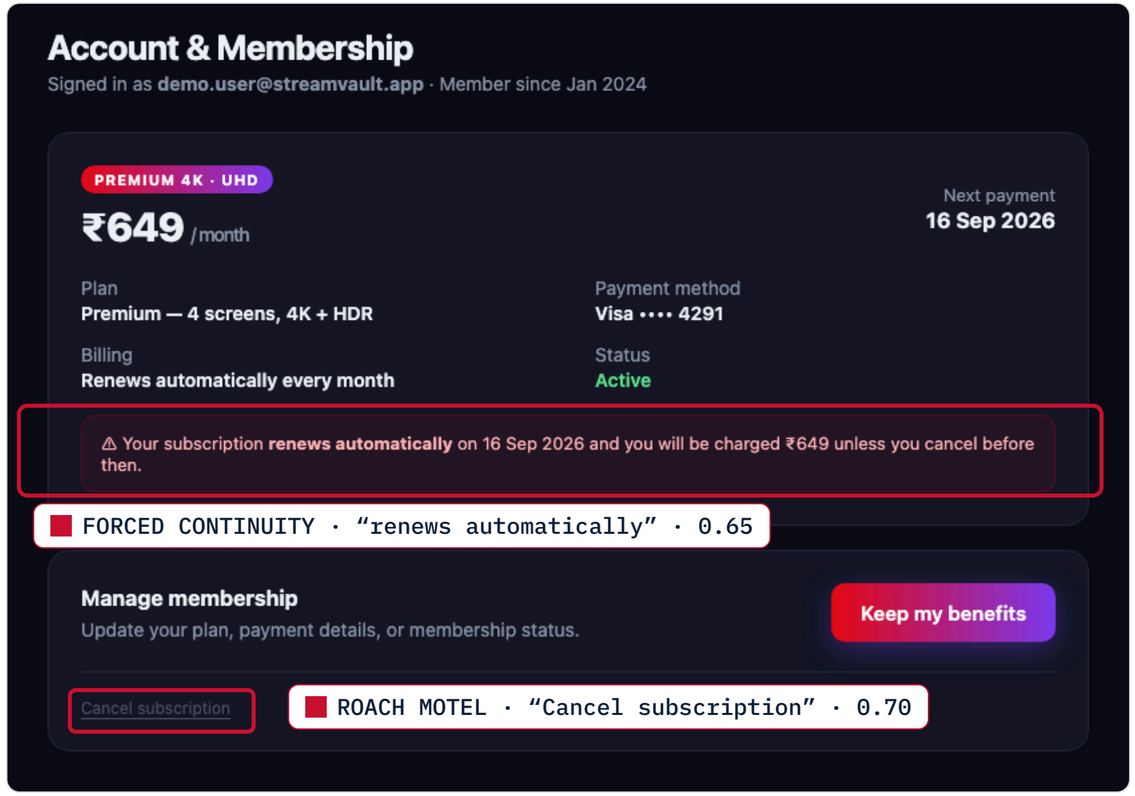
Problem Statement ID	FILL: PS ID from portal
Problem Statement Title	FILL: exact PS title from portal
Theme	FILL: exact theme string from portal
PS Category	**Software**
Team ID	FILL: Team ID
Team Name	FILL: team name as registered

TrapKiller

Find the trap. Quote the proof. Ask before the last click.

An AI agent that spots dark patterns on subscription pages, walks you out of the subscription, and waits for your approval before anything irreversible.

EXHIBIT A · REAL SCANNER OUTPUT



Built-in test site "StreamVault", made with the same tricks real subscription sites use.
Problem area: consumer protection · India's CCPA lists 13 banned dark patterns (2023).

TrapKiller: find the trap, quote the proof, ask before the last click

DETAILED EXPLANATION OF THE PROPOSED SOLUTION

- **Scanner:** reads a page and flags dark patterns with exact quotes.
- **Agent:** walks the cancel flow and skips “stay” offers.
- **Human gate:** the irreversible click waits for your approval.

HOW IT ADDRESSES THE PROBLEM

- Exposes tricks hidden in tiny links and guilt wording.
- Turns a maze of screens into one decision for the user.
- Leaves proof: a step-by-step log and a result screenshot.

INNOVATION AND UNIQUENESS OF THE SOLUTION

- **No quote, no finding:** AI claims are re-checked on the page.
- Own neural model catches reworded tricks (94.0% held-out).
- A knowledge graph learns routes and verdicts across cases.

TWO EXHIBITS, TWO VERDICTS — REAL PROTOTYPE OUTPUT

EXHIBIT A · SCAN (READ-ONLY) — HIGH RISK · 3 TRAPS

Confirmshaming	“We’d hate to see you go.”	0.77
Forced continuity	“renews automatically”	0.65
Roach motel	“Cancel subscription”	0.70

page: streamvault /account · every finding carries the words that prove it

EXHIBIT B · AGENT RUN (CANCELLATION)

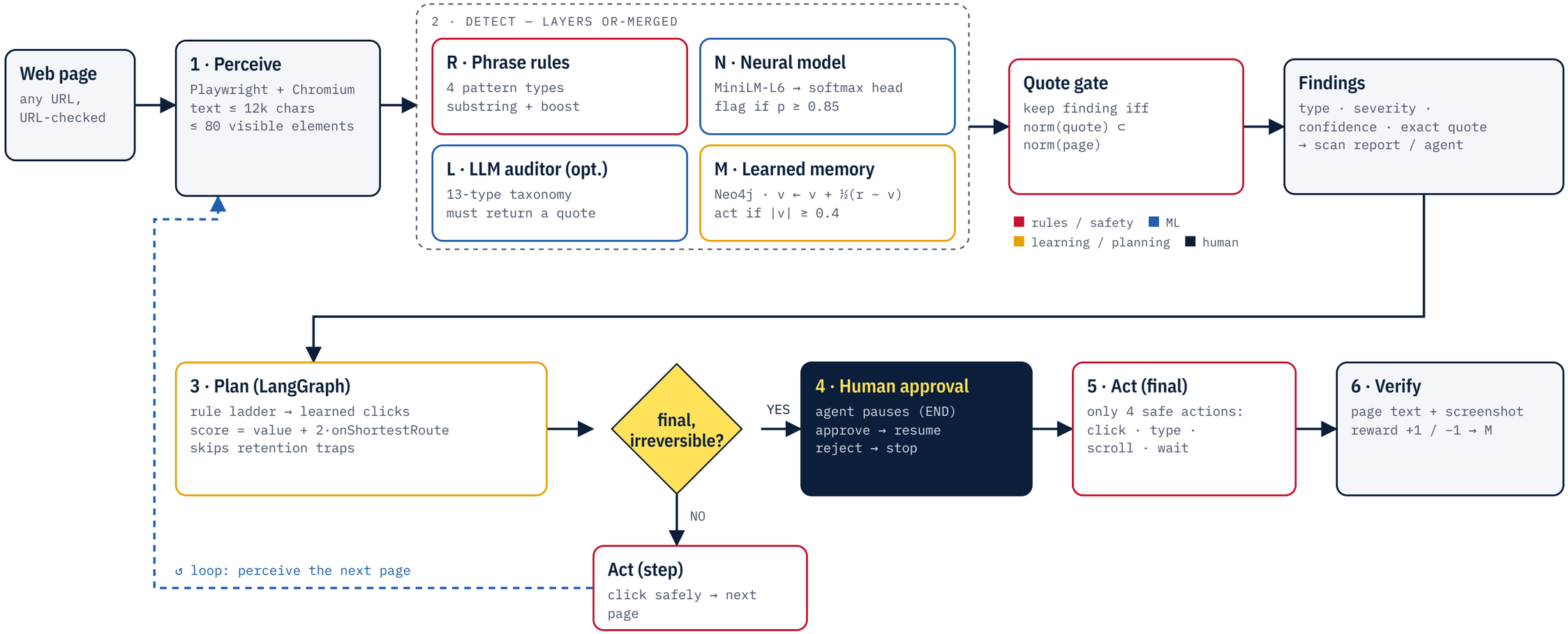


VERIFIED “Your subscription has been cancelled” · proof screenshot saved

Output from the TrapKiller prototype on its built-in StreamVault test site · Exhibit B: 2 clicks + 1 human approval, verified on the page

TRAPKILLER	CASE SHEET 2/6	REV A	TEAM [ID]

A quote-gated detector feeding a human-gated agent loop



STACK

Capture: Playwright · Chromium

Models: transformers.js (ONNX MiniLM) · LangGraph · LLM optional

Memory: Neo4j

App: Next.js 16 · React 19 · TypeScript

Tests: Vitest (74)

TRAPKILLER	CASE SHEET 3/6	REV A	TEAM [ID]

Every trap has a catching layer; every risk has a mechanism

TRAP → CATCHING LAYER

PATTERN	Rules	Neural	LLM*	Memory	Path map
Confirmshaming	●	○	○	○	–
Forced continuity	●	–	○	○	–
False urgency	●	●	○	○	–
Roach motel	●	–	○	○	●
Misdirection	–	●	○	○	–
Drip pricing / hidden costs	–	–	●	○	–
Nagging + 6 other types	–	–	●	○	–

● primary catcher · ○ backup · – none · *LLM layer is optional (API key); without it the rules + neural model + memory still run. Path map: counts the clicks to leave → trap score.

RISK → MITIGATION

The AI invents a trap	Quote gate drops any finding whose words are not on the page
Page text tries to steer the agent	Page text is data only; 4 safe actions; targets come from page text
Wrong irreversible click	Final click parked until a human approves; stop checked before every step
Unsafe target URL	Protocol, credential and private-IP checks with DNS re-check
False alarms	Neural cut-off p ≥ 0.85; a “false alarm” vote stops that wording
Site layout changes	Text-based perception + learned shortest route; LLM fallback

Technical

Open-source stack, CPU-only (23 MB neural model). The prototype runs end to end today; 74 automated tests pass.

Operational

AI flags and walks; the user approves the irreversible step. Every step is logged with a timestamp.

Economic

No licence costs. The LLM is optional; without it, running cost is one small server with a headless browser.

TRAPKILLER	CASE SHEET 4/6	REV A	TEAM [ID]

One decision instead of a maze — and evidence regulators can use

DECISION LEDGER · STREAMVAULT CANCEL FLOW

TODAY · BY HAND	▲ find the tiny grey cancel link	▲ resist “Keep my benefits”	▲ resist “I don’t want to lose my benefits”	■ confirm — no record kept	
WITH TRAPKILLER	◆ agent: flags 3 traps with quotes · skips both retention prompts · reaches the final step			**YOU: approve**	● verified + proof

Measured on the built-in test flow: 4 judgement calls under pressure → 1 informed decision. Real sites vary; this is the prototype’s own run, not a field study.

People

Stop paying for unwanted plans; leave without guilt or guesswork.

Regulators (CCPA)

Ready evidence: exact quotes, screenshots and per-site trap scores.

Honest businesses

Self-audit pages against the 13 banned patterns before launch.

Open ecosystem

A shared map of sites and tricks: each case helps the next user.

WHY IT MATTERS · SOURCED

75.7%
of 642 subscription sites & apps used a dark pattern [1]

67%
of 44,000+ Indian app users hit a subscription trap [4]

$2.5B
Amazon settlement over hard-to-cancel Prime [5]

USER STORY · ILLUSTRATIVE

Priya pays ₹649/month for a streaming plan she no longer uses. She pastes the account link. TrapKiller flags 3 traps, walks past two “stay” offers and pauses: **“Approve the final click?”** She taps approve; the case log and a screenshot are saved.

MEASURED

neural head

94.0%

held-out accuracy
(463 texts, public data)
74 / 74 tests pass

TRAPKILLER	CASE SHEET 5/6	REV A	TEAM [ID]

Built on published dark-pattern research and India's 2023 rules

HONEST BASELINE COMPARISON

CAPABILITY	By hand	Tracker apps	AI chatbot	TrapKiller
Spots dark patterns on the page	—	—	○	●
Quotes the exact words as proof	—	—	—	●
Walks the cancel steps for you	—	○	—	●
Asks before the irreversible click	○	○	—	●
Saves a proof screenshot + log	○	—	—	●
Learns from feedback across cases	—	—	—	●
Ranks sites by exit difficulty	—	—	—	●

● yes · ○ partly · — no. Columns are general categories of tools, not specific products. Rows ordered from basic to differentiating.

CODE & DEMO

github.com/Himanshusolanki24/
autonomous-dark-pattern-detection-and-consumer-protection-agent
live: autonomous-dark-pattern-detection-a.vercel.app

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